Prosoluble subgroups of the profinite completion of the fundamental group of compact 3-manifolds

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Abstract

We give a description of finitely generated prosoluble subgroups of the profinite completion of 3-manifold groups and toral relatively hyperbolic virtually compact special groups.

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1 Introduction

In recent years there has been a great deal of interest in detecting properties of the fundamental group $\pi_1(M)$ of a 3-manifold and other groups of geometric nature via its finite quotients, or more conceptually by its profinite completion (see [BRW17], [BMRS20], [CW22], [Wil18]). This motivates the study of the profinite

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completion $\widehat{\pi_1(M)}$ of the fundamental group of a 3-manifold. The study of $\widehat{\pi_1(M)}$ is based mainly on the work of Henry Wilton and the second author (see [WZ17]) where it is proved that the standard splittings of 3-manifold groups as a free product with amalgamation, HNN-extension, and generally as a graph of groups (we refer to them as free constructions later) survive after profinite completion of $\pi_1(M)$. Namely $\widehat{\pi_1(M)}$ admits the (profinite) Kneser–Milnor decomposition and then the (profinite) JSJ-decomposition, and for hyperbolic manifolds with cusps (after passing to a finite sheeted cover) there is a suitable hierarchy in which the corresponding actions on profinite trees are profinitely acylindrical.

At this point one can use the profinite analogue of Bass-Serre theory for groups acting on trees. Note, however, that this theory does not have the full strength of its classical original. The main theorem of Bass-Serre theory (that describes the subgroups of free constructions) asserts that a group G acting on a tree T is the fundamental group of a graph of groups $(\mathcal{G}, G\backslash T, D)$, where D is a maximal subtree of $G\backslash T$ and $\mathcal{G}$ consists of edge and vertex stabilizers of a connected transversal of $G\backslash T$. This does not hold in the profinite case. As a consequence, subgroup theorems do not hold, in general, for profinite groups and even for free profinite products, i.e., the profinite version of the Kurosh subgroup theorem is not valid. This motivates studying subgroup structure of free constructions for important subclasses of profinite groups. The most important subclasses of profinite groups are prosoluble and pro-p as they play the same role as finite soluble and finite p-groups in finite group theory.

Such a study for pro-p subgroups was performed in [WZ18], where the structure of finitely generated pro-p subgroups of profinite completions of the fundamental groups of 3-manifolds was described. The objective of this paper is to study prosoluble subgroups of $\widehat{\pi_1(M)}$.

We begin with the fundamental group of a compact hyperbolic 3-manifold.

Theorem A. *Let $\pi_1(M)$ be the fundamental group of a closed hyperbolic 3-manifold M and H a finitely generated prosoluble subgroup of the profinite completion of $\pi_1(M)$. Then H is projective, i.e., of cohomological dimension 1.*

Note that the Schreier theorem does not hold for profinite groups, i.e., subgroups of free profinite groups, in general, are not free but projective. So Theorem A tells that prosoluble subgroups of $\widehat{\pi_1(M)}$ are isomorphic to subgroups of a free profinite group.

The proof of Theorem A uses the dramatic developments of Agol ([Ago13]), Kahn–Markovic ([KM12]) and Wise ([Wis21]), whose works implies that the fundamental groups of closed hyperbolic 3-manifolds are also fundamental groups of compact, virtually special cube complexes (see [AFW15] for a summary of these developments); such groups are called virtually compact special. Virtually special groups originated in the work of Wise ([HW08], [HW12]). These groups have a central role in modern geometric group theory and Haglund and Wise showed that many hyperbolic groups are virtually special such as hyperbolic Coxeter groups and one relator groups with torsion.

The next theorem describes torsion-free finitely generated prosoluble subgroups of hyperbolic virtually special groups.

Theorem B. *Let G be a torsion-free, hyperbolic, virtually special group. Any finitely generated prosoluble subgroup H of the profinite completion of G is projective.*

Using that the standard cocompact arithmetic subgroups of $\mathrm{SO}(n,1)$ are virtually compact special ([BHW11, Theorem 1.10]) we deduce the following

Corollary. *Prosoluble subgroups of the profinite completion of standard cocompact arithmetic subgroups of* $\mathrm{SO}(n,1)$ *are projective.*

The next objective of this paper is to extend these results to relatively hyperbolic virtually compact special groups, in particular, to the fundamental hyperbolic 3-manifolds with cusps. In order to prove that, our arguments use both an analogue of Wise's Quasiconvex Hierarchy Theorem proved by Eduardo Einstein [Ein19] and the fact that it is preserved by the profinite completion established in [Zal23a].

Theorem C. *Let G be a torsion-free virtually compact special toral relatively hyperbolic group and H a finitely generated prosoluble subgroup of the profinite completion $\widehat{G}$ of G. Then one of the following statements holds:*

(i) H is virtually a subgroup of a free prosoluble product of abelian pro-p groups;

(ii) H is a virtually abelian group.

Using the [WZ18, Theorem 4.3] for 3-manifolds with cusps we deduce the following:

Theorem D. *Let M be a hyperbolic* 3*-manifold with cusps and H a finitely generated prosoluble subgroup of the profinite completion of $\pi_1(M)$. Then one of the following statements holds:*

(i) H is virtually a subgroup of a free prosoluble product of abelian pro-p groups;

(ii) H is a virtually abelian.

Finally, analyzing $\widehat{\pi_1(M)}$ for Thurston's geometries case-by-case, we obtain the classification of finitely generated prosoluble subgroups of $\widehat{\pi_1(M)}$ for an arbitrary compact, orientable 3-manifold.

Theorem E. *Let M be a compact orientable 3-manifold. If H is a finitely generated prosoluble subgroup of $\widehat{\pi_1(M)}$, then one of the following statements holds:*

(i) H is isomorphic to a subgroup of a free prosoluble product of pro-p groups from the following list of isomorphism types:

(1) For $p > 3$: C_p; $\mathbb{Z}_p$; $\mathbb{Z}_p \times \mathbb{Z}_p$; the pro-$p$ completion of $(\mathbb{Z} \times \mathbb{Z}) \rtimes \mathbb{Z}$ and the pro-p completion of a residually-p fundamental group of a non-compact Seifert fibred manifold with hyperbolic base of orbifold;

(2) For $p = 3$: in addition to the list of (1) we have a torsion-free extension of $\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3$ by C_3;

(3) For $p = 2$: in addition to the list of (1) we have C_{2^m}; D_{2^k}; Q_{2^n}; $\mathbb{Z}_2 \rtimes C_2$; the torsion-free extensions of $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ by one of C_2, C_4, C_8, D_2, D_4, D_8, Q_{16}; the pro-2 completion of the Klein-bottle group $\mathbb{Z} \rtimes \mathbb{Z}$; the pro-2 completion of all torsion-free extensions of a soluble group $(\mathbb{Z} \rtimes \mathbb{Z}) \rtimes \mathbb{Z}$ with a group of order at most 2;

(ii) $H \simeq \widehat{\mathbb{Z}}_\pi \rtimes C$ is a profinite Frobenius group where C is a finite cyclic group and π is set of primes;

(iii) H is a subgroup of a central extension of D_{2n}, a tetrahedral group T, a octahedral group O by a cyclic group of even order;

(iv) H is a subgroup of the profinite completion of a 3-dimensional Bieberbach group B (i.e, B is torsion-free virtually $\mathbb{Z}^3$);

(v) H is a subgroup of the profinite completion of a group extension of $\mathbb{Z}^2 \rtimes_A \mathbb{Z}$ by C where C is trivial or C_2 and $A \in \mathrm{GL}_2(\mathbb{Z})$ (and A is an Anosov matrix if C is trivial), or a central extension of $\mathbb{Z}$ by $\mathbb{Z}^2$;

(vi) H is an extension of a torsion-free procyclic group $\widehat{\mathbb{Z}}_\sigma$ by a subgroup H_0 of a free prosoluble product of finitely many finite cyclic p-groups with H_0 acting either trivially on $\widehat{\mathbb{Z}}_\sigma$ or by inversion.

In this theorem we are denoting by T the tetrahedral group of 12 rotational symmetries of a tetrahedron, by O the octahedral group of 24 rotational symmetries of a cube or of an octahedron. An Anosov matrix is a matrix $A \in \mathrm{GL}_2(\mathbb{Z})$ determined by an Anosov homeomorphism of the torus satisfying the following conditions $\mathrm{tr}(A) > 2$ if $\det(A) = 1$ and $\mathrm{tr}(A) \neq 0$ if $\det(A) = -1$ (see [GM23, Section 2]). We use the notation $\widehat{\mathbb{Z}}_\alpha = \prod_{p\in\alpha} \mathbb{Z}_p$ where α is a set of primes. Also, recall that a profinite Frobenius group $\widehat{\mathbb{Z}}_\pi \rtimes C$ is a group such that C is cyclic, $|C|$ divides $p-1$ for any $p \in \pi$ and $[c, z] \neq 1$ for all $1 \neq c \in C, 0 \neq z \in \widehat{\mathbb{Z}}_\pi$.

To prove Theorem E we make heavy use of the profinite version of Bass-Serre theory. In particular, we prove the following general theorem of independent interest:.

Theorem F. *Let $(\mathcal{G}, \Gamma)$ be an injective k-acylindrical finite graph of profinite groups, G its fundamental group. If H is a finitely generated prosoluble subgroup of G, then H embeds into a free profinite product $\coprod_{v\in\Gamma} H_v$, where $H_v = H \cap \mathcal{G}(v)^g$ and $g \in G$. Moreover, if $H \neq H_v$ for some v, then either H_v is finitely generated pro-p or $H \simeq \widehat{\mathbb{Z}}_\pi \rtimes C$ is a profinite Frobenius group.*

The celebrated Sela theorem (see [Sel97, Theorem 3.2]) states that a finitely generated group G is k-acylindrically accessible, i.e., if G is the fundamental group of a reduced k-acylindrical graph of groups $(\mathcal{G}, \Gamma)$, then Γ is finite and, in fact, bounded in terms of k and the minimal number of generators $d(G)$. This is equivalent to the statement that G acts on the corresponding Bass-Serre tree with bounded number of maximal vertex stabilizers up to conjugation. These statements are not equivalent, however, for profinite groups (one can see it using Proposition 2.15). We suggest that the correct profinite analog of accessibility is a bound for the number of maximal vertex stabilizers up to conjugation. Using Theorem F we prove an analogue of Sela's theorem for prosoluble groups.

Theorem G. *Let $(\mathcal{G}, \Gamma)$ be an injective k-acylindrical finite graph of profinite groups and G its profinite fundamental group. If H is a finitely generated prosoluble subgroup of G, then the number of maximal vertex stabilizers of H acting on the standard profinite Bass-Serre tree, up to conjugation, is at most $d(H)$.*

This paper is organized as follows. In Section 2 we introduce the basic ideas and notations of the profinite Bass-Serre theory as well as prove auxiliary results. In Section 3, we study pro-$\mathcal{C}$ groups acting k-acylindrically on a profinite simply connected graph. Closed pro-$\mathcal{C}$ subgroups of the profinite fundamental groups of a finite graph of profinite groups are studied in Section 4, where we prove Theorem G. We prove Theorem F in Section 5. Section 6 is dedicated to prosoluble subgroups of virtually special groups, where Theorems A, B, C and D are proved. We aplly them in Section 7 to the profinite completion of arithmetic hyperbolic manifold groups. In Section 8 we deal with Thurston's geometries of a compact 3-manifold in order to prove Theorem E.

When we deal with profinite groups our subgroups are assumed to be closed, homomorphisms to be continuous, finitely generated means topologically finitely generated. Throughout the paper we denote by $\mathcal{C}$ a class of finite groups closed for subgroups, quotients and extensions. We shall restate the results of the introduction inside of the section for easier reading.

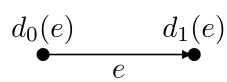
2 Finite graphs of pro-$\mathcal{C}$ groups and its fundamental group

In this section we will introduce the basic definitions and notations for profinite Bass-Serre theory used in this paper. This section is entirely based on [Rib17].

2.1 Finite graphs of pro-$\mathcal{C}$ groups

Definition 2.1 (Profinite graphs)**.** A *profinite graph* is a profinite space Γ with a distinguished non-empty subset $V(\Gamma)$ and two continuous maps $d_i : \Gamma \to V(\Gamma)$, $i = 0, 1$, which are the identity on $V(\Gamma)$.

The subset $V(\Gamma)$ is the *vertex-set* of Γ and $E(\Gamma) = \Gamma - V$ is the *edge-set* of Γ. If $e \in E(\Gamma)$, then we have



We call $d_0(e)$ and $d_1(e)$, respectively, by *initial* and *terminal* vertex. The maps d_0, d_1 are the *incidence maps.*

Definition 2.2. A *morphism* $\varphi : \Gamma \to \Delta$ of profinite graphs is a continuous map such that $\varphi d_i = d_i \varphi$ for $i = 0, 1$.

Definition 2.3. A profinite graph Γ is *connected* if every finite quotient graph of Γ is connected as an abstract graph.

Definition 2.4 (Simply connected profinite graphs)**.** Let G be a pro-$\mathcal{C}$ group acting freely on a profinite graph Γ. The natural epimorphism $\zeta : \Gamma \to G\backslash\Gamma$ is called a *Galois $\mathcal{C}$-covering* of $\Delta = G\backslash\Gamma$ and the associated group is denoted by $G = G(\zeta)$. If Γ does not have non-trivial Galois $\mathcal{C}$-coverings, then ζ is called *universal $\mathcal{C}$-covering* and, in this case, $\pi_1(\Gamma) = G(\Gamma|\Delta)$ is the *pro-$\mathcal{C}$ fundamental group* of Γ. If $\pi_1(\Gamma) = 1$ then we say that Γ is *$\mathcal{C}$-simply connected.*

Let Γ be a profinite graph. Define $E^*(\Gamma) = \Gamma\backslash V(\Gamma)$ and $\widehat{\mathbb{Z}}_{\widehat{\mathcal{C}}}$ as the pro-$\mathcal{C}$ completion of $\mathbb{Z}$.

Definition 2.5 (Profinite trees)**.** We say that Γ is a *pro-$\mathcal{C}$ tree* if the sequence

$$0 \longrightarrow [[\widehat{\mathbb{Z}}_{\widehat{\mathcal{C}}}(E^*(\Gamma), *)]] \xrightarrow{D} [[\widehat{\mathbb{Z}}_{\widehat{\mathcal{C}}}(V(\Gamma))]] \xrightarrow{\varepsilon} \widehat{\mathbb{Z}}_{\widehat{\mathcal{C}}} \longrightarrow 0$$

is exact, where $d(\bar{e}) = d_1(e) - d_0(e)$ for the image $\bar{e}$ of $e \in E(\Gamma)$ in $E^*(\Gamma)$ with $d(*) = 0$ and $\varepsilon(v) = 1$ for $v \in V(\Gamma)$.

In [Rib17, Section 3.10] we can see that "every simply connected profinite graph is a profinite tree", "if $\mathcal{C}$ is the variety of all finite soluble groups, $\mathcal{C}$-simply connected and $\mathcal{C}$-tree are equivalent".

We say that a profinite group G acts on the profinite tree T if it acts continuously on T and the action commutes with d_0 and d_1. We denote by G_t the stabilizer of $t \in T$ in G.

We define

$$T^g = \{m \in T : gm = m\}$$

to be the set of g-fixed points for a pro-$\mathcal{C}$ tree T, then T^g is again a $\mathcal{C}$-tree (see [Rib17, Theorem 4.1.5]). The subgroup of G generated by all vertex stabilizers usually will be denoted by $\widetilde{G}$.

Definition 2.6. Let Γ be a pro-$\mathcal{C}$ tree. Given the vertices $v, w \in \Gamma$ we can consider $[v, w]$ as the intersection of all pro-$\mathcal{C}$ subtrees of Γ containing v and w. We call $[v, w]$ a *geodesic*. The *length* $\ell([v, w])$ of $[v, w]$ is the number of edges in $[v, w]$.

Definition 2.7 (Finite graphs of profinite groups)**.** A *finite graph of pro-$\mathcal{C}$ groups* is a pair $(\mathcal{G}, \Gamma)$ where Γ is a connected finite graph, $\mathcal{G} = \dot{\bigcup}_{m \in \Gamma} \mathcal{G}(m)$ with $\mathcal{G}(m)$ a pro-$\mathcal{C}$ group for each m and there are continuous monomorphisms $\partial_i : \mathcal{G}(e) \to \mathcal{G}(d_i(e))$ for each edge $e \in E(\Gamma)$.

Definition 2.8. A *morphism*

$$\underline{\alpha} = (\alpha, \alpha') : (\mathcal{G}, \Gamma) \to (\mathcal{G}', \Gamma')$$

of graphs of pro-$\mathcal{C}$ groups consists of a pair of continuous maps $\alpha : \mathcal{G} \to \mathcal{G}'$ and $\alpha' : \Gamma \to \Gamma'$ such that the diagram

$$\begin{tikzcd} \mathcal{G} \arrow[r, "\alpha"] \arrow[d, "\pi"'] & \mathcal{G}' \arrow[d, "\pi'"] \\ \Gamma \arrow[r, "\alpha'"'] & \Gamma' \end{tikzcd}$$

is commutative. We say that $\underline{\alpha}$ is a *monomorphism* if α and α' are injective.

Definition 2.9. A finite graph of pro-$\mathcal{C}$ groups $(\mathcal{G}, \Gamma)$ is *reduced* if for every edge e which is not a loop (that is, $d_0(e) \neq d_1(e)$), neither $\partial_0 : \mathcal{G}(e) \to \mathcal{G}(d_0(e))$ nor $\partial_1 : \mathcal{G}(e) \to \mathcal{G}(d_1(e))$ is an isomorphism. We say that an edge e is fictitious if e is not a loop and ∂_0 or ∂_1 is an isomorphism.

Given a finite graph of groups with fictitious edges $e_1, ..., e_n$ we can collapse this edges transforming it in a reduced finite graph of groups with the same fundamental group (see [CZ23, Remark 2.8]).

2.2 The fundamental group of a finite graph of pro-$\mathcal{C}$ groups

For a totally new construction of the fundamental group of a profinite graph of pro-$\mathcal{C}$ groups using an approach similar to the abstract case, the reader should check [AZ22]. Here we will use the standard construction done in [Rib17].

Definition 2.10 (The fundamental group of a finite graph of profinite groups)**.** Let $(\mathcal{G}, \Gamma)$ be a finite graph of pro-$\mathcal{C}$ groups and T be a maximal subtree of Γ. *The pro-$\mathcal{C}$ fundamental group* of $(\mathcal{G}, \Gamma)$ is a group G with continuous homomorphisms

$$\nu_m : \mathcal{G}(m) \to G$$

and continuous maps $E(\Gamma) \to G$ denoted by $e \mapsto t_e$ such that $t_e = 1$ if $e \in E(T)$ and

$$(\nu_{d_0(e)}\partial_0)(x) = t_e(\nu_{d_1(e)}\partial_1)(x)t_e^{-1}$$

for all $x \in \mathcal{G}(e)$ and $e \in E(\Gamma)$ satisfying the following universal property: if H is a pro-$\mathcal{C}$ group, $\beta_m : \mathcal{G}(m) \to H$ are continuous homomorphisms, $e \to s_e$ ($e \in E(\Gamma)$) is a map with $s_e = 1$ if $e \in E(T)$ and

$$(\beta_{d_0(e)}\partial_0)(x) = t_e(\beta_{d_1(e)}\partial_1)(x)t_e^{-1}$$

for all $x \in \mathcal{G}(e)$ and $e \in E(\Gamma)$, then there is a unique continuous homomorphism $\delta : G \to H$ such that $\delta(t_e) = s_e$ and $\delta\nu_m = \beta_m$ for each $m \in \Gamma$.

We will denote the fundamental group of $(\mathcal{G}, \Gamma)$ by

$$G = \Pi_1(\mathcal{G}, \Gamma).$$

The reader can find proofs of existence and uniqueness of fundamental groups in [Rib17].

The basic free constructions such as free profinite groups, free profinite products, free profinite products with amalgamation and HNN-extensions are the most basic examples of fundamental groups of finite graphs of groups (see [Rib17, Section 6.2]).

The following lemma will be useful:

Lemma 2.11. *Let $(\mathcal{G}, \Gamma)$ be an injective profinite graph of pro-$\mathcal{C}$ groups and Δ a connected subgraph of Γ. Then $\Pi_1(\mathcal{G}, \Delta)$ is embedded in $\Pi_1(\mathcal{G}, \Gamma)$.*

Proof. First assume that $(\mathcal{G}, \Gamma)$ is a finite graph of finite groups. Note that $\Pi_1(\mathcal{G}, \Gamma)$ and $\Pi_1(\mathcal{G}, \Delta)$ are pro-$\mathcal{C}$ completions of the abstract fundamental groups ${\Pi_1}^{\text{abs}}(\mathcal{G}, \Gamma)$ and ${\Pi_1}^{\text{abs}}(\mathcal{G}, \Delta)$, respectively. Consider the collapse $\Gamma \to \Gamma/\Delta$ and the graph of groups $(\mathcal{G}_\Delta, \Gamma/\Delta)$ defining $\mathcal{G}_\Delta(m) = {\Pi_1}^{\text{abs}}(\mathcal{G}, \Delta)$ if m is the image of Δ and $\mathcal{G}_\Delta(m) = \mathcal{G}(m)$ for the others m. Since the edge groups of $(\mathcal{G}_\Delta, \Gamma/\Delta)$ are finite, the pro-$\mathcal{C}$ topology of ${\Pi_1}^{\text{abs}}(\mathcal{G}_\Delta, \Gamma/\Delta) = {\Pi_1}^{\text{abs}}(\mathcal{G}, \Gamma)$ induces the full pro-$\mathcal{C}$ topology on ${\Pi_1}^{\text{abs}}(\mathcal{G}, \Delta)$ (see [Rib17, Proposition 6.5.5]). Then by [Rib17, Proposition 6.5.3] the graph of groups $(\overline{\mathcal{G}}_\Delta, \Gamma/\Delta)$ of the pro-$\mathcal{C}$ completions of vertex and edge groups is injective. It follows that the pro-$\mathcal{C}$ completion of ${\Pi_1}^{\text{abs}}(\mathcal{G}, \Delta)$ embeds in the pro-$\mathcal{C}$ completion of ${\Pi_1}^{\text{abs}}(\mathcal{G}, \Gamma)$, as desired.

For the general case, given any open normal subgroup U of G, consider

$$\tilde{U} = \langle U \cap \mathcal{G}(v)^g : v \in V(\Gamma), g \in G \rangle.$$

We have (see [AZ22, Proposition 2.15 and Theorem 3.9])

$$G_U = G/\tilde{U} = \Pi_1(\mathcal{G}_U, \Gamma)$$

such that

$$\Pi_1(\mathcal{G}, \Gamma) = \varprojlim \Pi_1(\mathcal{G}_U, \Gamma)$$

and

$$\Pi_1(\mathcal{G}, \Delta) = \varprojlim \Pi_1(\mathcal{G}_U, \Delta).$$

Since the diagram

$$\begin{tikzcd} \Pi_1(\mathcal{G}, \Gamma) \arrow[r] \arrow[d] & \Pi_1(\mathcal{G}, \Delta) \arrow[d] \\ \Pi_1(\mathcal{G}_U, \Gamma) \arrow[r, hook] & \Pi_1(\mathcal{G}_U, \Delta) \end{tikzcd}$$

commutes, we get $\Pi_1(\mathcal{G}, \Delta) \leqslant \Pi_1(\mathcal{G}, \Gamma)$. $\square$

With standard arguments one can deduce the following:

Proposition 2.12. *Let G be the fundamental group of a finite reduced graph of profinite groups with at least one edge. If G is not virtually cyclic, then G contains a non-abelian free profinite group. In particular, if $\mathcal{C}$ does not contains all finite groups, then G is not pro-$\mathcal{C}$.*

Proof. Let U be an open normal subgroup and $\tilde{U}$ as defined in the previous lemma. Then $G_U = G/\tilde{U}$ is the fundamental group $\Pi_1(\mathcal{G}_U, \Gamma)$ of a finite graph $(\mathcal{G}_U, \Gamma)$ of finite groups $\mathcal{G}(m)\tilde{U}/\tilde{U}$ with ∂_i induced by ∂_i of $(\mathcal{G}, \Gamma)$ (note that the order of $\mathcal{G}(m)\tilde{U}/\tilde{U}$ is bounded by the order of $\mathcal{G}(m)/(\mathcal{G}(m) \cap U)$ which is finite since U is open in G). As $G = \varprojlim_U G_U$, $G/\tilde{U}$ is not virtually cyclic for some U and also we can choose such U such that $(\mathcal{G}_U, \Gamma)$ is reduced. But G_U is the profinite completion of the abstract fundamental group ${\Pi_1}^{\mathrm{abs}}(\mathcal{G}_U, \Gamma)$ (cf. [Rib17, Proposition 6.5.1]) and ${\Pi_1}^{\mathrm{abs}}(\mathcal{G}_U, \Gamma)$ is not virtually cyclic, so ${\Pi_1}^{\mathrm{abs}}(\mathcal{G}_U, \Gamma)$ contains a non-abelian free subgroup of finite index and therefore $G/\tilde{U}$ contains a non-abelian open free profinite group. Then G contains a non-abelian free profinite group and so, cannot be pro-$\mathcal{C}$. □

Definition 2.13 (Standard graph)**.** Given a finite graph of pro-$\mathcal{C}$ groups $(\mathcal{G}, \Gamma)$ with fundamental group G, the corresponding *standard graph* is

$$S = S(G) = \dot{\bigcup_{m \in \Gamma}} G/\mathcal{G}(m).$$

The vertices of S are the cosets $g\mathcal{G}(v)$ with $v \in V(\Gamma)$ and $g \in G$; the edges of S are the cosets $g\mathcal{G}(e)$ with $e \in E(\Gamma)$ and $g \in G$; the incidence maps of S are given by

$$d_0(g\mathcal{G}(e)) = g\mathcal{G}(d_0(e)), \quad d_1(g\mathcal{G}(e)) = gt_e\mathcal{G}(d_1(e))$$

with $e \in E(\Gamma)$ and $t_e = 1$ if $e \in E(T)$.

The fact that S is indeed a profinite graph as well as the basic properties of S are explained in [Rib17, Section 6.3]. In particular, it is proved in [Rib17, Theorem 6.3.5] that S is $\mathcal{C}$-simply connected and so is a pro-$\mathcal{C}$ tree. Besides, G acts continuously on S with $G \backslash S \simeq \Gamma$.

2.3 Auxiliary Results

In this subsection we will prove some facts relevant to the general theory of profinite groups acting on profinite graphs. The results here will be helpful to show that k-acylindrical actions are well-behaved under taking quotients in the next section.

Lemma 2.14. *Let G be a profinite group acting on a simply connected profinite graph Γ and Σ a finite connected subgraph of Γ. Then $H = \langle G_v \mid v \in V(\Sigma)\rangle$ is the fundamental group of a finite tree of pro-$\mathcal{C}$ groups $\Pi_1(\mathcal{H}, \Delta)$ whose edge and vertex groups are stabilizers of the corresponding edges and vertices of a connected transversal of $\Delta = H\backslash H\Sigma$ in $T = H\Sigma$.*

Proof. Put $T = H\Sigma$. Since the quotient of T modulo every open subgroup U of H is connected, T is connected. By [Rib17, Proposition 3.7.3], Σ is simply connected. By [Rib17, Proposition 3.9.2], $\Delta = H\backslash T$ is simply connected and so is a finite tree. By [Rib17, Theorem 6.6.1], $H = \Pi_1(\mathcal{H}, \Delta)$ whose edge and vertex groups are stabilizers of the corresponding edges and vertices of a connected transversal of Δ in T. □

Proposition 2.15. *Let G be a profinite group acting on a simply connected profinite graph Γ and let e be an edge of S. Suppose the stabilizers of the vertices v, w of the edge e satisfy the following condition: $|G_v : G_e| + |G_w : G_e| > 4$. Then the group $H = \langle G_v, G_w\rangle$ is an amalgamated free profinite product $G_v \amalg_{G_e} G_w$. In particular, if $\mathcal{C}$ does not contains all finite groups, then H is not pro-$\mathcal{C}$.*

Proof. By Lemma 2.14

$$H = \Pi_1(\mathcal{H}, \Delta) \simeq G_v \amalg_{G_e} G_w.$$

Then, by Proposition 2.12, H contains a non-abelian free profinite group and so cannot be pro-$\mathcal{C}$. □

The previous result does not hold if there is some e such that $|G_v : G_e| = |G_w : G_e| = 2$. When it occurs we will say that $\langle G_v, G_e, G_w\rangle$ is of *dihedral type* or that the edge e *generates a dihedral type group*.

Corollary 2.16. *Let $\mathcal{C}$ be a variety not containing all finite groups and G a pro-$\mathcal{C}$ group acting on a simply connected profinite graph Γ. Suppose that G has no edges e generating a dihedral type group. Then for every edge e with vertices v, w and non-trivial edge stabilizer, we have $G_v = G_e$ or $G_w = G_e$.*

Proof. Suppose that, for a non-trivial edge stabilizer e with vertices v and w, we have G_v and G_w both different from G_e. By hypothesis, the index of G_e in G_v or in G_w is greater than 2. By Proposition 2.15, $H = \langle G_v, G_w\rangle$ is a not a pro-$\mathcal{C}$ subgroup of G, a contradiction. □

Lemma 2.17. *Let $G = G_1 \amalg_H G_2$ be a splitting of a profinite group G as an amalgamated free profinite product of profinite groups G_1, G_2 and $H_1 \leqslant G_1, H_2 \leqslant G_2$ be subgroups such that $H_1 \cap H \leqslant U \geqslant H \cap H_2$ for some open normal subgroup U of G. Then $\langle H_1, H_2\rangle = L_1 \amalg_K L_2$ with $L_1 \leqslant H_1U, L_2 \leqslant H_2U, K \leqslant H \cap U$.*

Proof. By [ZM89a, Proposition 4.4], $\langle H_1, H_2\rangle = L_1 \amalg_K L_2$ with $L_1 \leqslant G_1$, $L_2 \leqslant G_2$, $K \leqslant H$. Recall that the fundamental group $\Pi_1(\mathcal{G}, \Gamma)$ of a finite graph of profinite groups is a completion of the abstract fundamental group ${\Pi_1}^{\text{abs}}(\mathcal{G}, \Gamma)$ of the same graph of groups with respect to to the topology generated by the finite index normal subgroups N of ${\Pi_1}^{\text{abs}}(\mathcal{G}, \Gamma)$ such that $N \cap \mathcal{G}(v)$ is open in $\mathcal{G}(v)$. Note that G is the fundamental group of a finite graph of profinite groups. We can consider $U \cap G_1$ and $U \cap G_2$ to apply [CZ23, Lemma 3.1] on $G^{\text{abs}} = G_1 *_H G_2$ to deduce that $\langle H_1, H_2\rangle^{\text{abs}} = L_1 *_K L_2$ where $L_1 \leqslant H_1(U \cap G_1)$, $L_2 \leqslant H_2(U \cap G_2)$ and $K \leqslant H \cap U$. Then $\langle H_1, H_2\rangle = \overline{L}_1 \amalg_K \overline{L}_2$ with $\overline{L}_1 \leqslant H_1U$, $\overline{L}_2 \leqslant H_2U$ and $K \leqslant H \cap U$. $\square$

Corollary 2.18. *Let $G = G_1 \amalg_H G_2$ be a splitting of a profinite group G as an amalgamated free profinite product of profinite groups G_1, G_2 and $H_1 \leqslant G_1, H_2 \leqslant G_2$ be subgroups such that $H_1 \cap H = 1 = H_2 \cap H$. Then $\langle H_1, H_2\rangle = H_1 \amalg H_2$.*

Proof. Since we have $H_1 \cap H = 1 = H_2 \cap H$ we deduce from Lemma 2.17 that $K \leqslant U$, $L_1 \leqslant H_1U$, $L_2 \leqslant H_2U$ for an arbitrary open normal subgroup U. Since the intersection of all such U is trivial, $L_1 = H_1$, $L_2 = H_2$, $K = 1$ and the result follows. $\square$

Let G be a pro-$\mathcal{C}$ group acting on a pro-$\mathcal{C}$ tree T. Recall that, for any subgroup U of G, we set

$$\widetilde{U} = \langle U \cap G_v : v \in V(T)\rangle.$$

Proposition 2.19. *Let G be a pro-$\mathcal{C}$ group acting on a pro-$\mathcal{C}$ tree T and U an open normal subgroup of G. Then $D_U = \{m \in \widetilde{U}\backslash T : (G/\widetilde{U})_m \neq 1\}$ is a profinite subgraph of $\widetilde{U}\backslash T$.*

Proof. Put $G_U = G/\widetilde{U}$, $T_U = \widetilde{U}\backslash T$. It suffices to prove that D_U is closed in T_U. As $\{(G_U)_m : m \in T\}$ is a continuous family (see [Rib17, Lemma 5.2.2]), the set $\{m \in \widetilde{U}\backslash T : (G/\widetilde{U})_m = 1\} = \{m \in \widetilde{U}\backslash T : (G/\widetilde{U})_m \leqslant U/\widetilde{U}\}$ is open. Hence $D_U = \{m \in \widetilde{U}\backslash T : (G/\widetilde{U})_m \neq 1\}$ is closed. $\square$

2.4 Relatively projective groups

As was already mentioned in the introduction, the profinite analogue of the Kurosh Subgroup Theorem does not hold for free profinite products of profinite groups and so the algebraic structure of closed subgroups of free profinite products is not clear. To compensate this, Dan Haran (see [Har87]) introduced the notion of relatively projective profinite groups and proved that a subgroup H of a free profinite product $\coprod_{x\in X} G_x$ is projective relatively to the family $\{H \cap G_x^g \mid x \in X, g \in$

$G\}$. We shall use a slightly more restricted notion here from [HZ23] as it better fits into the profinite version of Bass-Serre theory and suffices for our purpose.

Definition 2.20. A *group-pile* is a pair $\mathbf{G} = (G, T)$ consisting of a profinite group G, a profinite space T and a continuous action of G on T. We say that $\mathbf{G}$ is *finite* if both G and T are finite.

Definition 2.21. A *morphism* of group-piles $\alpha : (G, T) \to (G', T')$ consists of a group homomorphism $\alpha : G \to G'$ and a continuous map $\alpha : T \to T'$ such that $\alpha(gt) = \alpha(g)\alpha(t)$ for all $t \in T$ and $g \in G$. The above morphism is an *epimorphism* if $\alpha(G) = G'$, $\alpha(T) = T'$ and for every $t' \in T'$ there is $t \in T$ such that $\alpha(t) = t'$ and $\alpha(G_t) = {G'}_{t'}$. It is *rigid* if α maps G_t isomorphically onto ${G'}_{\alpha(t)}$, for every $t \in T$, and the induced map of the orbit spaces $G\backslash T \to G'\backslash T'$ is a homeomorphism.

Definition 2.22. An *embedding problem* for a group-pile $\mathbf{G}$ is a pair (φ, α) where $\varphi : \mathbf{G} \to \mathbf{H}$ and $\alpha : \mathbf{G'} \to \mathbf{H}$ are morphisms of group-piles such that α is an epimorphism. It is *finite* if $\mathbf{G'}$ is finite. It is *rigid* if α is rigid. A *solution* to the embedding problem is a morphism $\gamma : \mathbf{G} \to \mathbf{G'}$ such that $\alpha \circ \gamma = \varphi$.

It means that the diagram

$$\begin{tikzcd} & (G,T) \arrow[dl, dashed, "\gamma"'] \arrow[d, "\varphi"] \\ (G',T') \arrow[r, "\alpha"'] & (H,X) \end{tikzcd}$$

is commutative.

For the next definition, let $\mathcal{G}$ be a continuous family of subgroups of G and $\mathcal{C}$ a class of finite groups closed for subgroups, quotients and extensions.

Definition 2.23. A pile $\mathbf{G}$ is *$\mathcal{C}$-projective* if every finite rigid $\mathcal{C}$-embedding problem for $\mathbf{G}$ has a solution. A group G is *$\mathcal{G}$-projective* if every finite $\mathcal{C}$-embedding problem has a weak solution, that is, there exists a group homomorphism $\gamma : G \to G'$ such that $\alpha \circ \gamma = \varphi$.

We will make use of the following result:

Theorem 2.24. *[Zal23b, Theorem 4.1]*

(i) Let G be a prosoluble group acting on a profinite tree D with trivial edge stabilizers. Then $(G, V(D))$ is $\mathcal{S}$-projective, where $\mathcal{S}$ is the class of all finite soluble groups.

(ii) Let $\mathcal{C}$ be a class of finite groups closed for subgroups, quotients and extensions. Let (G,T) be a projective pro-$\mathcal{C}$ pile. Suppose that exists a continuous section $\sigma : G\backslash T \to T$. Then G acts on a pro-$\mathcal{C}$ tree with trivial edge stabilizers and non-trivial vertex stabilizers being stabilizers of points in T.

We also need the following proposition:

Proposition 2.25. *[Zal23b, Proposition 2.5] Let $\mathcal{C}$ be a class of finite groups closed for subgroups, quotients and extensions. Let (G,T) be a $\mathcal{C}$-projective group-pile such that there exists a continuous section $\sigma : G\backslash T \to T$. Then G embeds into a free pro-$\mathcal{C}$ product*

$$\coprod_{s\in Im(\sigma)} G_s \amalg F$$

where F is a free pro-$\mathcal{C}$ group of rank $d(G)$, the minimal number of generators of G. Moreover, every G_t is of the form $G_s^g \cap G$ for some $g \in \coprod_{s\in Im(\sigma)} G_s \amalg F$, $s \in Im(\sigma)$.

3 Acylindrical actions and prosoluble groups

In this section we prove a generalization of [WZ17, Lemma 11.2]. This result allows us to deduce that if $\mathcal{C}$ does not contain all finite groups and if G is a pro-$\mathcal{C}$ group acting k-acylindrically on a simply connected profinite graph T, then for any closed normal subgroup N generated by vertex stabilizers the quotient group G/N acts k-acylindrically on $N\backslash T$.

Definition 3.1. An action of a profinite group G on a profinite graph T is called *k-acylindrical* if the stabilizer of any geodesic in T of length greater than k is trivial.

This definition is sometimes stated in the following way:

Definition 3.2. An action of a profinite group G on a profinite tree T is called *k-acylindrical* if, for every nontrivial $g \in G$, the subtree of fixed points

$$T^g = \{m \in T : gm = m\}.$$

has diameter at most k.

It is easy to see that both definitions are equivalent: suppose that Definition 3.1 holds and there is a path $[v,w]$ with length greater than k in T^g for some nontrivial $g \in G$. Then $g \in G_{[v,w]}$, a contradiction. Conversely, if T^g has diameter at most k for every nontrivial $g \in G$ and $[v,w]$ is a geodesic of length greater than k. If $g \in G_{[v,w]}$, then $gv = v$ and $gw = w$ so that $[v,w] \in T^g$, a contradiction.

Definition 3.3. We say that a finite graph of profinite groups $(\mathcal{G}, \Gamma)$ is *k-acylindrical* if its profinite fundamental group acts k-acylindrically on its standard profinite tree.

Proposition 3.4. *Let $(\mathcal{G}, \Gamma)$ be a k-acylindrical finite graph of profinite groups and G its fundamental group. Then there is no edge e such that*

$$|G_{d_0(e)} : G_e| = 2 = |G_{d_1(e)} : G_e|$$

with $G_e \neq 1$.

Proof. Assume that there is at least one edge e with

$$|G_{d_0(e)} : G_e| = 2 = |G_{d_1(e)} : G_e|$$

and $G_e \neq 1$. Put $v = d_0(e)$, $w = d_1(e)$ and set $H = \langle G_v, G_w \rangle$. Let $D = H(v \cup e \cup w)$. Since for every open subgroup U of H the quotient of $U \backslash D$ is connected, so is D. By Lemma 2.14 $H = G_v \amalg_{G_e} G_w$. Since G_e is normal in H, by [Rib17, Proposition 4.2.2, (b)], G_e acts trivially on D, contradicting the k-acylindricity of the action. This finishes the proof. □

This proposition shows that if the action is k-acylindrical, then we can apply the results in Subsection 2.3 without any restrictions.

Recall that a profinite graph Γ is also an abstract graph. Let G be a profinite group acting on it. Put $D = \{e \cup d_0(e) \cup d_1(e) \mid G_e \neq 1\}$. Then D is an abstract subgraph of Γ and is maximal among abstract subgraphs of Γ having non-trivial edge stabilizers for all its edges.

Theorem 3.5. *Suppose $\mathcal{C}$ does not contain all finite groups and let G be a pro-$\mathcal{C}$ group acting k-acylindrically on a simply connected profinite graph Γ. Then the maximal abstract subgraph D of Γ such that each $e \in D$ satisfies $G_e \neq 1$ has finite diameter $\leqslant 2k$. Moreover, for every connected component D^* of D there at most one other connected component at a finite distance to D^* in which case stabilizers of edges and vertices of D^* are of order 2.*

Proof. Suppose on the contrary there exists a finite connected subgraph Σ of D whose diameter is $> 2k$. Let $H = \langle G_v \mid v \in V(\Sigma) \rangle$ and $T = H\Sigma$. By Lemma 2.14 and [Rib17, Proposition 4.1.1], H is the fundamental group of a tree of pro-$\mathcal{C}$ groups $\Pi_1(\mathcal{H}, H \backslash T)$ whose edge and vertex groups are stabilizers of the corresponding edges and vertices of a connected transversal of $H \backslash T$ in T. Hence, by Proposition 3.4 there is no edge e such that

$$|\mathcal{H}(d_0(e)) : \mathcal{H}(e)| = 2 = |\mathcal{H}(d_1(e)) : \mathcal{H}(e)|$$

with $G_e \neq 1$. Let $(\mathcal{H}_{red}, H\backslash T_{red})$ be a reduced tree of groups obtained from $(\mathcal{H}, H\backslash T)$. If it has more than one vertex, then H cannot be pro-$\mathcal{C}$ by Proposition 2.12. Hence, $(\mathcal{H}_{red}, H\backslash T_{red})$ has one vertex v only which implies that H fixes a vertex w in T and, in fact, $w \in \Sigma$. As the diameter of Σ is greater than $2k$, it follows that there exists a geodesic $[u, w]$ in Σ of length greater than k which is fixed by $H_u \neq 1$ (see [Rib17, Corollary 4.1.6]) contradicting the k-acylindricity of the action. This proves the first part of the statement.

Now if there exist a connected component D_0 at finite distance from D^* and $v \in D^*, w \in D_0$ are vertices with $G_v \ncong C_2$, then $\langle G_v, G_w\rangle$ is not virtually cyclic and so by Proposition 2.12 can not be pro-$\mathcal{C}$, a contradiction. Similarly, if there exists still another connected component D_1 at finite distance from D^* and $u \in V(D_1)$, then $\langle G_v, G_u, G_w\rangle$ is not virtually cyclic again, contradicting Proposition 2.12 again. This finishes the proof. □

Corollary 3.6. *Suppose we are in the situation of Theorem 3.5 and Γ has infinite diameter. Then Γ possesses an edge e with $G_e = 1$, namely $G_e = 1$ for any $e \notin D$.*

Corollary 3.7. *Suppose that $G\backslash D$ has finitely many connected components. Then G acts on a simply connected profinite G-quotient graph T of Γ with trivial edge stabilizers and vertex stabilizers equal to the stabilizers of some vertices Γ. Moreover, the number of maximal (by inclusion) G-stabilizers of vertices in T, up to conjugation, equals to the number of maximal G-stabilizers of vertices in Γ and this number is bounded by the number of connected components of $G\backslash D$.*

Proof. By [CZ22, Lemma 3.2] the closure of any connected component $\overline{D^*}$ is a profinite subgraph of Γ of diameter $\leqslant 2k$, hence $\overline{D} = \bigcup_i \overline{D_i^*}$ has finite diameter, where ${D_i}^*$ runs via orbit representatives of the connected components of D. Moreover, as every connected component $\overline{D^*}$ of $\overline{D}$ has finite diameter, its stabilizer $\mathrm{Stab}(\overline{D^*})$ fixes some vertex $v_* \in V(\overline{D^*})$ (cf. [Ser02, Corollary, page 20]). Now we can collapse all connected components of the profinite subgraph $\overline{D}$ of Γ (see [Rib17, Exercise 2.1.11]; the resulting profinite quotient graph T of Γ is G-invariant and by [Rib17, Proposition 3.9.1] is simply connected. The edge stabilizers of T are trivial; the vertices of T whose stabilizers are non-trivial are exactly the vertices to which the connected components of D were collapsed; thus the only non-trivial vertex stabilizers of T are stabilizers of the connected component $\mathrm{Stab}(\overline{D}_*) = G_{v_*}$ of $\overline{D}$. This concludes the proof. □

Theorem 3.8. *Suppose $\mathcal{C}$ does not contain all finite groups and let G be a pro-$\mathcal{C}$ group acting k-acylindrically on a simply connected profinite graph Γ and U a closed normal subgroup of G. Then the action of $G_U = G/\tilde{U}$ on $\tilde{U}\backslash\Gamma$ is k-acylindrical.*

Proof. By Theorem 3.5 an abstract subgraph D whose edge stabilizers are non-trivial has diameter at most k. Consider the natural action of G_U on $\Gamma_U = \tilde{U}\backslash\Gamma$. We shall prove that the same is true in $\Gamma_U = \tilde{U}\backslash\Gamma$; we use the subscript U for images in $\tilde{U}\backslash\Gamma$.

Let D be a maximal connected abstract subgraph of Γ where each edge $e \in D$ satisfies $G_e \neq 1$. Note that any edge e_U not in D_U is an image of an edge $e \notin D$ and since G_e is trivial, $(G_U)_{e_U}$ is trivial. Therefore, a maximal connected (abstract) subgraph of Γ_U whose all edge stabilizers are non-trivial is contained in D_U and so has diameter at most k, which implies that the action of G_U on Γ_U is k-acylindrical. □

Recall that for any open normal subgroup U of a pro-$\mathcal{C}$ group G acting on a pro-$\mathcal{C}$ tree T we set

$$\widetilde{U} = \langle U \cap G_v : v \in V(T)\rangle,$$

and $G_U = G/\widetilde{U}$, $T_U = \widetilde{U}\backslash T$.

Proposition 3.9. *Suppose $\mathcal{C}$ does not contain all finite groups and let G be a pro-$\mathcal{C}$ group acting k-acylindrically on a simply connected profinite tree T. The group G_U acts on a profinite simply connected G_U-quotient tree T_U^* of T_U with trivial edge stabilizers and vertex stabilizers equal to the maximal (by inclusion) stabilizers of vertices of T_U. Moreover, if G is prosoluble then $(G_U, V(T_U^*))$ is $\mathcal{S}$-projective, where $\mathcal{S}$ is the class of all finite soluble groups.*

Proof. By Proposition 2.19 $D_U = \{m \in \widetilde{U}\backslash T \mid (G/\widetilde{U})_m \neq 1\}$ is a profinite subgraph of T_U. We can collapse all connected components of the profinite subgraph D_U of T_U (see [Rib17, Exercise 2.1.11]); the resulting profinite quotient graph T_U^* of T_U is G-invariant and by [Rib17, Proposition 3.9.1] is simply connected. The edge stabilizers of T_U^* are trivial; the vertices of T_U^* whose stabilizers are non-trivial are exactly the vertices to which the connected components of D_U were collapsed; thus the only non-trivial vertex stabilizers of T_U^* are stabilizers of the connected component $\mathrm{Stab}(D_U)_*$ of D_U. But as was explained in the proof of Theorem 3.8 any connected component of D_U has diameter $\leqslant 2k$. Hence $\mathrm{Stab}(D_U)_*$ stabilizers a vertex in $(D_U)_*$ (cf. [Ser02, Corollary, Page 20]) and clearly the stabilizer of this vertex is maximal by inclusion. This concludes the first part of the proof.

Suppose now G is prosoluble. By item (i) of Theorem 2.24 $(H_U, V(T_U^*))$ is $\mathcal{S}$-projective that finishes the proof of second statement. □

Lemma 3.10. *Let G be a prosoluble group acting k-acylindrically on a simply connected profinite tree T. Suppose for some primes $p \neq q$ and some $v, w \in V(T)$ from different G-orbits, the stabilizer G_v has a subgroup of order p and the stabilizer G_w has a subgroup of order q as subquotients. Then there exists a subgroup H of*

G that admits an epimorphism $f : H \to C_{pq}$ to a cyclic group of order pq such that $|f(H_v)| = p, |f(H_w)| = q$ and $K = \ker(f)$ is not generated by its vertex stabilizers. Moreover, non-trivial subgroups of the images of H_v and H_w in $H/\widetilde{K}$ do not commute up to conjugation.

Proof. W.l.o.g we may assume that G_v and G_w are maximal (by inclusion) vertex stabilizers. Choose an open normal subgroup U of G such that G_vU/U and G_wU/U have subgroups G_p of order p and G_q of order q, respectively. Consider the action of $G_U = G/\widetilde{U}$ on $T_U = \widetilde{U}\backslash S$ and note that in G_U the vertex stabilizers of the images v_u, w_u of v and w in T_U are G_vU/U and G_wU/U (still in different orbits) and refining U we may assume that G_vU/U and G_wU/U are maximal vertex stabilizers. By Proposition 3.9 G_U acts on a simply connected (infinite) profinite tree T_U^* with trivial edge stabilizers, the vertex stabilizers of T_U^* are maximal vertex stabilizers of T_U and $(G_U, V(T_U^*))$ is $\mathcal{S}$-projective, where $\mathcal{S}$ is the class of all finite soluble groups. Hence G_p and G_q are subgroups of some stabilizers of vertices of T_U^* from distinct G_U-orbits. Note that if a subgroup of G generated by its vertex stabilizers then its image in G_U generated by its vertex stabilizers as well, and vertex stabilizers in G_U are finite. Therefore to prove the lemma it suffices to construct a subgroup H in G_U containing G_p and G_q and an epimorphism $f : H \to C_{pq}$ to a cyclic group of order pq with $|f(G_p)| = p, |f(G_q)| = q$ such that $\ker(f)$ is not generated by torsion.

Put $H = \langle G_p, G_q\rangle$. Then by [Zal24, Lemma 4.8] setting

$$\mathcal{H} = \{H \cap (G_U)_v \mid v \in V(T_U)\}$$

there is a profinite H-space X such that

(a) $\mathcal{H} = \{(H)_x \mid x \in X\}$;

(b) $(H)_x \cap (H)_{x'} = 1$ for $x \neq x'$;

(c) $\mathbf{H} = (H, X)$ is a $\mathcal{C}$-projective pile.

Moreover, since H is finitely generated, it is second countable and so by [Zal24, Remark 4.9] there exists a second countable such X. Therefore there exists a continuous section $\sigma : H\backslash X \to X$ (see [RZ10, Lemma 5.6.7]).

Hence by Proposition 2.25 H embeds into a free prosoluble product $H_0 = \coprod_{s\in Im(\sigma)} H_s \amalg F$, where F is free prosoluble, such that each $H_x = H_s^h$ for some $h \in H_0$. Choose $x_1, x_2 \in X$ such that H_{x_1} contains G_p and H_{x_2} contains G_q, observe that x_1, x_2 are from different H-orbits and let $s_1 \neq s_2 \in Im(\sigma)$ such that H_{s_1} and H_{s_2} conjugate to H_{x_1} and H_{x_2} respectively.

By [Rib17, Theorem 5.3.4] H_0 is an inverse limit of free prosoluble products with finitely many factors which are quotients free factors of H_0 and so there exists such a free prosoluble product $\coprod_{i=1}^{n} H_{s_i} \amalg F$ with finitely many factors, where H_{s_1} and H_{s_2} are still the free factors. Then factoring out all factors but H_{s_1} and H_{s_2} we arrive at a free prosoluble product $H_{s_1} \amalg H_{s_2}$, where H_{s_1} and H_{s_2} are still conjugate to the images of H_{x_1} and H_{x_2} respectively.

Now define a natural epimorphism $H_{s_1} \amalg H_{s_2} \to H_{s_1} \times H_{s_2}$ that sends H_{s_1} onto H_{s_1} and H_{s_2} onto H_{s_2} identically. The kernel of this natural epimorphism is free prosoluble (by [RZ10, Theorem 9.1.6]). Hence the kernel of f factors through the kernel of the restriction of this epimorphism to the image M_U of H_U in $H_{s_1} \amalg H_{s_2}$ which is torsion-free. Therefore the kernel of f is not generated by torsion as required. Moreover, as non-trivial subgroups of H_{s_1}, H_{s_2} in $H_{s_1} \amalg H_{s_2}$ do not commute up to conjugation (cf. [Rib17, Theorem 9.1.2]), hence non-trivial subgroups of H_{x_1} and H_{x_2} of M_U do not commute up to conjugation as well. □

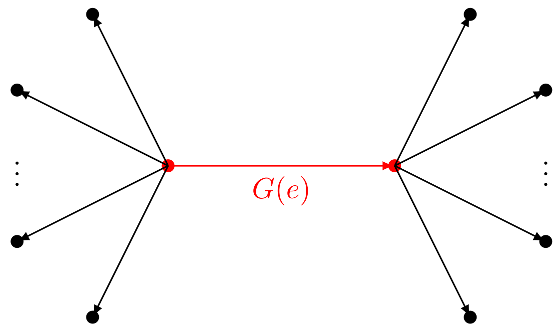
4 Closed subgroups of fundamental groups of finite graphs of profinite groups with finite edge groups

In this section we will conclude an important step to prove Theorem E. The main theorem of this section gives us necessary conditions for a prosoluble group to be a subgroup of the fundamental group of an injective k-acylindrical finite graph of profinite groups with finite edge groups. The fundamental groups of a finite graph of profinite groups are always profinite (or pro-$\mathcal{C}$) in this section unless otherwise said; if vertex groups are finitely generated they are just profinite completion of the classical Bass-Serre fundamental groups.

This lemma is a general version of [CZ23, Proposition 2.16]:

Proposition 4.1. *Let G be the fundamental pro-$\mathcal{C}$ group of an injective reduced finite graph of pro-$\mathcal{C}$ groups $(\mathcal{G}, \Gamma)$. Suppose that there is an edge e of Γ such that $G(e) = 1$ and $G(d_i(e)) \neq 1$ for $i = 0, 1$. Then G splits as a free pro-$\mathcal{C}$ product of either fundamental groups of two subgraphs of groups of $(\mathcal{G}, \Gamma)$ or as the fundamental group of the subgraph of groups of $(\mathcal{G}, \Gamma)$ and $\widehat{\mathbb{Z}}_{\mathcal{C}}$.*

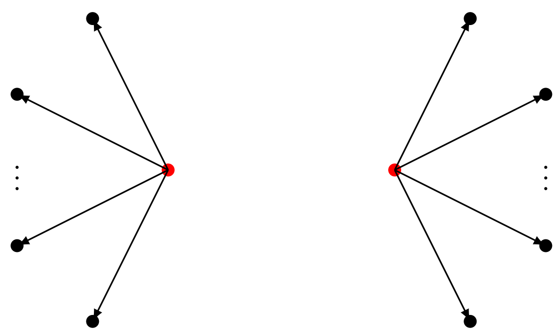
Proof. Let



be a slice of the graph Γ.

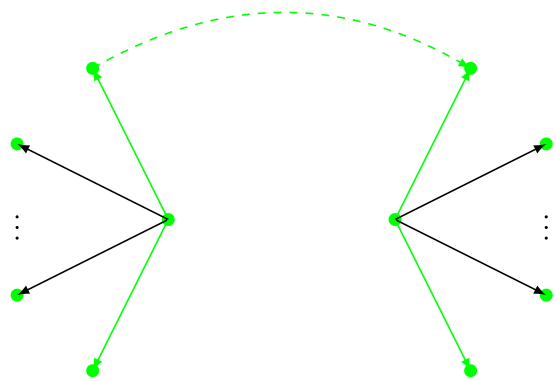
After removing the edge e we have two possibilities: either the resulting graph is connected or not connected. We shall prove that G splits as a free profinite product in each of the possibilities.

Case 1. After removing the edge e the graph Γ become not connected:

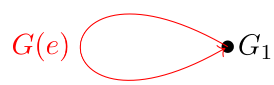


Let Γ_1, Γ_2 be the connected components and G_1, G_2 its fundamental groups, respectively. They are embedded in G by Lemma 2.11. It follows that $G = G_1 \amalg G_2$.

Case 2. After removing e the graph remains connected



then we can choose a maximal subtree T not containing e and use it to calculate the fundamental group G_1 of the graph of groups restricted to $\Gamma - \{e\}$. So, G is the fundamental group of the loop with edge correspondent to $G(e)$ and vertex correspondent to G_1 so that $G = \mathrm{HNN}(G_1, G(e), t) = G_1 \amalg \langle t \rangle$, since $G(e) = 1$.



$\square$

Theorem 4.2. *Let $(\mathcal{G}, \Gamma)$ be an injective k-acylindrical finite graph of profinite groups with finite edge groups and G its profinite fundamental group. Suppose $\mathcal{C}$ does not contain all finite groups. If H is a pro-$\mathcal{C}$ subgroup of G, then there exists $e \in E(\Gamma)$ and a conjugate H_0 of H such that $H_0 \cap G(e) = 1$ and for any such e*

$$H_0 \leqslant U_1 \amalg U_2$$

where $U_1 \simeq \Pi_1(\mathcal{U}, \Delta)$ is the profinite fundamental group of an injective k-acylindrical finite graph of profinite groups with finite edge groups, Δ a connected component of $\Gamma \backslash \{e\}$ and U_2 is either isomorphic to $\widehat{\mathbb{Z}}$ or similarly is the profinite fundamental group of an injective k-acylindrical finite graph of profinite groups with finite edge groups.

Proof. Consider the action of H on the standard profinite tree $S = S(G)$. By Corollary 3.6 there is an edge e of S for which H_e is trivial. Then there is an open subgroup U_H of G containing H such that $(U_H)_e = 1$. By [ZM89a, Corollary 4.5] U_H is the profinite fundamental group of a finite graph of groups $(\mathcal{U}, \Delta)$ whose vertex and edge groups are U_H-stabilizers of vertices and edges of S. Thus one of the edge groups $U(\bar{e})$ of $(\mathcal{U}, \Delta)$ is trivial. Assuming, w.l.o.g., that $(\mathcal{U}, \Delta)$ is reduced we can apply Proposition 4.1 to deduce that U_H splits as a free profinite product. As $U(\bar{e})$ is conjugate by some element $g \in G$ into some edge group of $(\mathcal{G}, \Gamma)$ we may conjugate H and U by this element to get the required H_0. This finishes the proof. ☐

The second author [Zal24, Theorem 1.1] described the prosoluble subgroups of free profinite products. Using this we can deduce the following

Corollary 4.3. *One of the following holds for H_0 if it is prosoluble:*

(i) *There is a prime p such that $H_0 \cap uU_iu^{-1}$ is pro-p for every $i = 1, 2$ and $u \in U_1 \amalg U_2$; besides if H is finitely generated then $\sum_{i,u} d(H_0 \cap uU_iu^{-1}) \leqslant d(H)$.*

(ii) *$H_0 \simeq \widehat{\mathbb{Z}}_\pi \rtimes C$ is a profinite Frobenius group.*

(iii) *A conjugate of H_0 is a subgroup of U_i for some $i = 1, 2$.*

If H is torsion-free, then item (ii) of Corollary 4.3 does not occur.

It is not difficult to construct an example satisfying Theorem 4.2.

Example 4.4. Let G_1, G_2 be profinite Frobenius groups (or any finite Frobenius groups) with isomorphic complement C. Consider the group $G = G_1 \amalg_C G_2$. Then C is malnormal in G and so G satisfies the hypothesis of Theorem 4.2. Then any prosoluble subgroup H has the structure stated in Corollary 4.3.

The next proposition is of independent interest.

Proposition 4.5. *Let $G = \Pi_1(\mathcal{G}, \Gamma)$ be the profinite fundamental group of a k-acylindrical finite graph of profinite groups. Let $v, w \in V(\Gamma)$ be vertices at distance at least $2k + 1$. Then $\langle G(v), G(w) \rangle = G(v) \amalg G(w)$.*

Proof. Let $[v, w]$ be a shortest path between v and w. Let $G(v, w)$ be the fundamental group of the finite graph of profinite groups restricted to $[v, w]$. By Lemma 2.11 $G(v, w)$ is a subgroup of G generated by vertex groups of $[v, w]$. Let e be an edge of $[v, w]$ at distance $> k$ from w and v. Then $G(v) \cap G(e) = 1 = G(w) \cap G(e)$. Note that $G(v, w)$ splits over $G(e)$ as a free amalgamated profinite product $G(v,w) = G_1 \amalg_{G(e)} G_2$, where G_1, G_2 are profinite groups generated by vertex groups of the connected components of $[v,w]\backslash\{e\}$ (see Lemma 2.11), so that $G(v) \leqslant G_1$ and $G(w) \leqslant G_2$. By Corollary 2.18, $\langle G(v), G(w)\rangle = G(v) \amalg G(w)$ as required. $\square$

For the proof of the next theorem we shall need two simple lemmas.

Lemma 4.6. *Let G be a pro-$\mathcal{C}$ group acting on a pro-$\mathcal{C}$ tree T. Let $\mathcal{U} = \{U_i\}$ be a filtered from below family of subgroups of G with $H = \bigcap_{U\in\mathcal{U}} U$. Then $\widetilde{H} = \bigcap_{U\in\mathcal{U}} \widetilde{U}$.*

Proof. By [Rib17, Proposition 4.1.1] $\widetilde{U}\backslash T$ is a pro-$\mathcal{C}$ tree and so

$$\left(\bigcap_{U\in\mathcal{U}} \widetilde{U}\right)\backslash T = \varprojlim_{U\in\mathcal{U}} \widetilde{U}\backslash T$$

is a pro-$\mathcal{C}$ tree (cf. [Rib17, Proof of Lemma 4.2.7]). Hence by [Rib17, Proof of Lemma 4.2.7] $\bigcap_{U\in\mathcal{U}} \widetilde{U}$ is generated by the vertex stabilizers and so equal to $\widetilde{H}$. $\square$

Lemma 4.7. *Let G be an extension of an abstract free group F by a group C_{pq} of order pq for primes $p \neq q$. Suppose that G contains finite subgroups A and B of order p and q that do not commute up to conjugation. Then the natural epimorphism $f : G \to C_{pq}$ with kernel F factors through an abstract free product $C_p * C_q$.*

Proof. By the results of Karrass, Pietrowski, Solitar, Cohen and Scott [KPS73, Coh06, Sco74]) G splits as the fundamental group ${\Pi_1}^{\mathrm{abs}}(\mathcal{G},\Gamma)$ of a finite graph of finite groups and assuming w.l.o.g. that $(\mathcal{G},\Gamma)$ is reduced, one sees that the vertex groups of $(\mathcal{G},\Gamma)$ are of order at most pq and edge groups are of order at most p or q.

As any finite group is conjugate into a vertex group, we shall assume, w.l.o.g., that A and B are in the vertex groups of $(\mathcal{G},\Gamma)$. Let $(\mathcal{G},\Gamma_A)$ be the maximal subgraph of groups of $(\mathcal{G},\Gamma)$ containing A in its every vertex group and $(\mathcal{G},\Gamma_B)$ is the maximal subgraph of groups of $(\mathcal{G},\Gamma)$ containing B in its every vertex group. Since A and B do not commute up to conjugation, Γ_A and Γ_B do not intersect. Then factoring out the normal closure of all subgroups of order q contained in the vertex groups of $(\mathcal{G},\Gamma_A)$ and all subgroups of order p contained in the vertex groups of $(\mathcal{G},\Gamma_B)$ we obtain the fundamental group ${\Pi_1}^{\mathrm{abs}}(\mathcal{H},\Gamma)$, where A is still contained in a vertex group of $(\mathcal{H},\Gamma_A)$, B is still contained in the vertex groups of $(\mathcal{H},\Gamma_B)$ but neighboring edge groups of Γ_A and Γ_B are trivial. This means that A and B are free factors of ${\Pi_1}^{\mathrm{abs}}(\mathcal{H},\Gamma)$ and so so the group ${\Pi_1}^{\mathrm{abs}}(\mathcal{H},\Gamma)$ can be mapped to $A * B$. $\square$

Theorem 4.8. *Let $(\mathcal{G}, \Gamma)$ be a k-acylindrical finite graph of profinite groups, G its profinite fundamental group and $S = S(G)$ its standard profinite tree. Let H be a subgroup of G such that there exist some non-trivial non-conjugate vertex stabilizers H_v, H_w for $v, w \in S$ which are not both pro-p for any prime p. Then H is not prosoluble.*

Proof. Suppose on the contrary H is prosoluble. Since H_v and H_w are not both pro-p for a fixed prime p, there exist primes $p \neq q$ such that groups of order p and q are subquotients of H_v and H_w respectively. Then by Corollary 3.10 there exists an open subgroup L of H that admits an epimorphism $f_L : L \to C_{pq}$ to the cyclic group of order pq with the kernel not generated by vertex stabilizers such that $f_L(L_v)$ has order p and $f_L(L_w)$ has order q. Thus, w.l.o.g., we may assume that $H = L$ and $f_L = f_H$. By [RZ10, Lemma 8.3.8] f_H extends to an epimorphism $f_U : U \to C_{pq}$ from some open subgroup U of G containing H and by Lemma 4.6 we can choose such U with the kernel K of f_U not generated by its vertex stabilizers. Let $\widetilde{K}$ be the normal subgroup of K generated by the vertex stabilizers. Then $\widetilde{K}$ is normal in U and we can consider $U/\widetilde{K}$ acting on $\widetilde{K}\backslash S$ which is simply connected by [ZM89a, Lemma 4.7]. Then by [ZM89a, Proposition 4.4] $U/\widetilde{K}$ is the fundamental profinite group of a finite graph of finite groups $\Pi_1(\mathcal{U}, U\backslash S)$ and so is the profinite completion of the abstract fundamental group $\Pi = \Pi_1^{abs}(\mathcal{U}, U\backslash S)$ (cf. [Rib17, Proposition 6.5.6]). Note that we have an induced epimorphism $f_K : U/\widetilde{K} \to C_{pq}$ and we denote by f^{abs} its restriction to Π. Since $K/\widetilde{K}$ is projective (see [Rib17, Corollary 4.1.3] or [ZM89b, Theorem 2.6]) and so is torsion free, the kernel $\ker(f^{abs})$ is free and so by Lemma 4.7 f^{abs} factors through the free product $C_p * C_q$. Hence f_K factors through the free profinite product $C_p \amalg C_q = \widehat{C_p * C_q}$. Thus we have the following commutative diagram:

$$\begin{tikzcd}
 & U \arrow[d] & \\
H \arrow[ur] \arrow[r] \arrow[dr, dashed] \arrow[ddr, "f_H"'] & U/\widetilde{K} \arrow[d] & \Pi \arrow[l] \arrow[d] \\
 & C_p \amalg C_q \arrow[d] & C_p * C_q \arrow[l] \arrow[dl] \\
 & C_{pq} &
\end{tikzcd}$$

Since $f_H(H_v)$ has order p and $f_H(H_w)$ has order q, it follows from this commutative diagram that the image of H in $C_p \amalg C_q$ contains a group of order p and a group of order q.

Choose now elements $c_p, c_q \in S_n$, $n > 4$ of order p and q, such that any conjugate of them generate non-soluble subgroups (see [Jar94, Lemmas 1.1 and 1.2]) and let $f_p : C_p \amalg C_q \to C_p < S_n$, $f_q : C_p \amalg C_q \to C_q < S_n$ be the corresponding epimorphisms. Then we have an epimorphism $\varphi : C_p \amalg C_q \to S_n$. Since the image of H in $C_p \amalg C_q$ contains the group of order p and the group of order q, it maps epimorphically to S_n which is not soluble, a contradiction. This finishes the proof. $\square$

Corollary 4.9. *Let $(\mathcal{G}, \Gamma)$ be a k-acylindrical finite graph of profinite groups and G its profinite fundamental group. Let H be a prosoluble subgroup of G. Then one of the following holds:*

(i) all non-trivial intersections $H \cap G(v)^g$, $g \in G, v \in V(\Gamma)$ are pro-p;

(ii) $H \simeq \widehat{\mathbb{Z}}_\pi \rtimes C$ is a profinite Frobenius group;

(iii) $H \leqslant G(v)^g$ for some $g \in G, v \in V(\Gamma)$.

Proof. Suppose that the non-trivial $H \cap G(v)^g$ are not all pro-p and that there is no $g \in G, v \in V(\Gamma)$ such that $H \leqslant G(v)^g$. Then by Theorem 4.8 all $H \cap G(v)^g$ are conjugate in H. Consider the action of H on the standard profinite tree $S = S(G)$. Then $H \cap G(v)^g$ are exactly the stabilizers of vertices of S in H. Thus all vertex stabilizers of H are conjugate in H. It follows that $D = \{m \in S \mid H_m \neq 1\}$ has only one connected component up to translation and so by Corollary 3.7 we can collapse the connected components of $\bar{D}$ to obtain the action of H on the simply connected profinite graph T with trivial edge stabilizers. Let U be an open normal subgroup of H. Then by Theorem 3.8 $H_U = H/\tilde{U}$ acts k-acylindrically on $T_U = \tilde{U}\backslash T$ with finite vertex stabilizers and note that we still have that all vertex stabilizers are conjugate in H_U and true for any subgroup of H_U whose vertex groups are not all pro-p. Then $(H_U)_v$ is finite soluble, and not a finite p-group for almost all U. So we can choose in $(H_U)_v$ a subgroup of the form $C = A \rtimes C_l$, a semidirect product of elementary abelian p-group A and a cyclic group of order l for some distinct primes p, l. Let F_U be a projective open normal subgroup of H_U and consider $F_U C = F_U \rtimes C$. Since C is self-normalized in $F_U C$ (cf. [Rib17, Corollary 7.1.6 (a)]) it satisfies [Zal24, Lemma 2.4] and so applying it we deduce that H_U is soluble. Then by [Rib17, Theorem 4.2.11] H_U is Frobenius. Note that $H = \varprojlim_U H_U$ and so by [RZ10, Theorem 4.6.9] H is profinite Frobenius as well. $\square$

Proposition 4.10. *Let $(\mathcal{G}, \Gamma)$ be an injective k-acylindrical finite graph of profinite groups, G its profinite fundamental group, S its standard graph and H is a prosoluble subgroup of G. If H is finitely generated, then it has only finitely many maximal vertex stabilizers up to conjugation. In particular, if D is the maximal* *abstract subgraph of S such that each $e \in D$ satisfies $H_e \neq 1$, then D has finitely many connected components up to translation.*

Proof. By Corollary 4.9 we have that H satisfies (i), (ii) or (iii).

If H satisfies (ii) and not (i) and (iii), then all vertex stabilizers are finite and conjugate (as they are Hall subgroups, cf. [RZ10, Theorem 2.3.5]).

It remains to analyze (i) and (iii). Choosing any open subgroup U of H, we can pass to the quotient $H_U = H/\tilde{U}$ acting on S_U. Recall that $H = \varprojlim_U H_U$ and $S = \varprojlim_U S_U$. So, to prove that the number of maximal vertex stabilizers of H is finite, up to conjugation, it is enough to prove that the number of maximal vertex stabilizers up to conjugation in H_U is bounded independently on U.

Suppose first that H satisfies (i). Then by Proposition 3.9 and Theorem 3.8 H_U acts acylindrically on a simply connected profinite tree S_U^* with trivial edge stabilizers, vertex stabilizers equal to the maximal stabilizers of S_U and $(H_U, V(\mathcal{S}_U^*))$ is $\mathcal{S}$-projective. Moreover, by Theorem 3.5 the connected components of $D_U = \{e \cup d_0(e) \cup d_1(e) \mid (H_U)_e \neq 1\}$ have finite diameter and so S_U^* is non-trivial by Corollary 3.6. By [Zal24, Proposition 4.9], the number of non-conjugated $(H_U)_{v_*}$, $v_* \in S_U^*$, is bounded by $d(H_U) \leqslant d(H)$. Since the non-trivial vertex stabilizers of S_U^* are stabilizers of connected components of D_U, it follows that the number of connected components of D_U and hence the number of the maximal vertex stabilizers of S_U is bounded by $d(H)$ up to conjugation, independently on U.

Suppose that H satisfies (iii). If $H \leqslant G(v)^g$ then H fixes a vertex and the same is true for each H_U. It follows that the number of maximal vertex stabilizers, up to conjugation, is trivially bounded by $d(H)$.

Thus, in both cases (i) and (iii) each H_U has finitely many maximal vertex stabilizers, up to conjugation. This finishes the proof. □

Proof of Theorem G follows from Proposition 4.10.

5 The finitely generated case

In this section we shall study finitely generated prosoluble subgroups of profinite groups acting k-acylindrically on profinite trees.

Theorem F. *Let $(\mathcal{G}, \Gamma)$ be an injective k-acylindrical finite graph of profinite group and G its fundamental group. If H is a finitely generated prosoluble subgroup of G, then H embeds into a free profinite product $\coprod_{v \in V(\Gamma)} H_v$ of finitely many finitely* *generated profinite groups $H_v = H \cap G(v)^g$ for some $g \in G$. Moreover, if $H \neq H_v$ for some v, then either H_v is finitely generated pro-p or $H \simeq \widehat{\mathbb{Z}}_\pi \rtimes C$ is a profinite Frobenius group.*

Proof. Note that the abstract subgraph $D = \{m \mid H_m \neq 1\}$ has finite diameter by Theorem 3.5, and by Proposition 4.10 D has finitely many connected components up to translation. Therefore $H\backslash D$ has finitely many connected components and so by Corollary 3.7 there exists a simply connected H-quotient graph $\overline{S}$ on which H acts with trivial edge stabilizers.

By item (i) of Theorem 2.24, $(H, V(\overline{S}))$ is an $\mathcal{S}$-projective group-pile where $\mathcal{S}$ is the class of all finite soluble groups. By [Zal24, Remark 4.9] there exists a second countable space T with the same non-trivial point stabilizers such that (H, T) is a projective group-pile. Since T is second countable, the quotient map $T \to H\backslash T$ has a continuous section (see [RZ10, Lemma 5.6.7] or [Rib17, 1.3.10]). By Theorem 2.24 (H, T) is an $\mathcal{S}$-projective group-pile such that the quotient map $T \to H\backslash T$ has a continuous section. Therefore by Proposition 2.25 H embeds into a free prosoluble product

$$\coprod_{t \in Im(\sigma)}^{\mathcal{S}} H_t \amalg^{\mathcal{S}} F$$

where F is a free prosoluble group. Moreover, every H_t is of the form $H \cap G(v)^g$. By Corollary 4.9, the H_t are either all pro-p, or H is Frobenius and all H_t are conjugate as they are Hall subgroups of H. Then by [Zal24, Theorem 1.4] H embeds into a desired free profinite product. □

Corollary 5.1. *Let $(\mathcal{G}, \Gamma)$ be an injective k-acylindrical finite graph of profinite groups, G its fundamental group and S its standard profinite graph. If H is a finitely generated prosoluble subgroup of G, then it has only finitely many vertex stabilizers up to conjugation. Moreover, $\sum_v d(H_v) \leqslant d(H)$, where v runs through representatives of H-orbits in S.*

Proof. Follows from Theorem F and [Zal24, Corollary 1.6]. □

6 Hyperbolic virtually special groups

In this section we apply the previous results to hyperbolic and relatively hyperbolic virtually compact special groups.

We start with some definitions.

Definition 6.1. A group is *virtually special* if there is a finite index subgroup isomorphic to the fundamental group of a special cube complex. A group is *virtually compact special* if there is a finite index subgroup isomorphic to the fundamental group of a compact special cube complex whose hyperplanes satisfy some extra conditions [HW08, Definition 3.2].

Definition 6.2. A family $\{H_1, ..., H_n\}$ of subgroups of G is *malnormal* if, for any $g \in G$, $H_i \cap H_j^g \neq 1$ implies $i = j$ and $g \in H_i$.

Definition 6.3. A *hierarchy of groups of length* 0 is a single vertex labeled by a group. A *hierarchy of groups of length* n is a graph of groups $(\mathcal{G}_n, \Gamma_n)$ together with hierarchies of length $n-1$ on each vertex of Γ_n (cf. [WZ17, Definition 1.3]).

In [WZ17, Theorem 1.4] it is shown that a hyperbolic group G is virtually compact special if, and only if, G has a subgroup Γ_0 of finite index with a malnormal hierarchy. It is also shown that, in this case, Γ_0 has a 1-acylindrical action on a profinite tree (see [WZ17, Lemma 7.3]). In the relatively hyperbolic virtually compact special case, Γ_0, in addition, has a malnormal hierarchy terminating in parabolic subgroups (see [Zal23a, Theorem 3.2]).

Theorem B. *Let G be a torsion-free, hyperbolic, virtually compact special group. Any finitely generated prosoluble subgroup H of the profinite completion of G is projective.*

Proof. Let G be a hyperbolic virtually compact special group and H a finitely generated prosoluble subgroup of $\widehat{G}$. Let $H_0 = \widehat{\Gamma}_0 \cap H$ where Γ_0 is the subgroup defined above. We use induction on the length of the malnormal hierarchy described above. Since $\widehat{\Gamma_0}$ is a fundamental group of a graph of profinite groups $(\mathcal{G}, \Gamma)$ whose action on the standard graph is 1-acylindrical (see [WZ17, Lemma 7.3]), by Theorem F, we have an embedding

$$H_0 \hookrightarrow \coprod_{v \in V(\Gamma)} H_0 \cap G(v)^{g_i}, g_i \in \widehat{\Gamma}_0.$$

Then by Corollary 4.3 either $H_0 \leqslant H_v^g$ or all $H_0 \cap G(v)^g$ are pro-p. In the first case the result follows immediately from the induction hypothesis. In the second case we observe first that, by Theorem F, $H_0 \cap G(v)^g$ are finitely generated which allows us to deduce that $H_0 \cap G(v)^g$ are free pro-p (cf. [WZ17, Theorem F]). It follows that H_0 is projective and so H is virtually projective. Since it is torsion-free, it is projective (see [Zal04, Proposition 2.5]). □

Theorem B yields a description of prosoluble subgroups of closed hyperbolic 3-manifolds.

Theorem A. *Let $\pi_1(M)$ be the fundamental group of a closed hyperbolic 3-manifold M and H a finitely generated prosoluble subgroup of the profinite completion of $\pi_1(M)$. Then H is projective.*

Proof. By [AFW15, Theorem 5.4] $\pi_1(M)$ is virtually compact special. Hence the result follows from Theorem B. □

Let G be a residually finite group hyperbolic relative to a malnormal family of parabolic subgroups $\mathcal{P} = \{P_1, ..., P_n\}$. Any subgroup of a conjugate of P_i will be called *parabolic*. A closed subgroup of the profinite completion of a maximal parabolic subgroup of G be called *parabolic subgroup* of $\widehat{G}$.

Proposition 6.4. *Let G be a relatively hyperbolic virtually compact special group and H a finitely generated prosoluble subgroup of the profinite completion of G. Then there is an open subgroup H_0 of H such that one of the following statements holds:*

(i) H_0 is isomorphic to a subgroup of a free prosoluble product of pro-p subgroups of parabolic subgroups,

(ii) H_0 is isomorphic to a subgroup of a parabolic subgroup.

Proof. Let G be a relatively hyperbolic virtually compact special group and H a finitely generated prosoluble subgroup of $\widehat{G}$. In [Zal23a, Proof of Theorem 1.3] we see that there is a subgroup Γ_0 such that $\widehat{\Gamma_0}$ acts 1-acylindrically on a profinite tree. Let $H_0 = \widehat{\Gamma}_0 \cap H$. As in the proof of Theorem B, by Theorem F, we have an embedding

$$H_0 \hookrightarrow \coprod H_0 \cap G(v)^g.$$

We assume w.l.o.g that H_0 is torsion-free. Then by Corollary 4.3 either $H_0 \leqslant H_v^g$ or all $H_0 \cap G(v)^g$ are pro-p. In the first case the result follows immediately from the induction hypothesis.

In the second case we observe first that by Proposition 4.10 $H_0 \cap G(v)^g$ are finitely generated which allows to apply [Zal23a, Theorem 1.4] to deduce that $H_0 \cap G(v)^g$ are free pro-p products of parabolic pro-p groups and a free pro-p group. □

Definition 6.5. We say that a relatively hyperbolic group is *toral* if it is torsion-free hyperbolic relative to a set $\{A_1, ..., A_n\}$ of parabolic subgroups where each A_i is a finitely generated abelian group.

Now we are in the position deduce from Proposition 6.4

Theorem C. *Let G be a torsion-free virtually compact special toral relatively hyperbolic group and H a finitely generated prosoluble subgroup of the profinite completion $\widehat{G}$ of G. Then one of the following statements holds:*

(i) H is virtually a subgroup of a free prosoluble product of abelian pro-p groups;

(ii) H is a virtually abelian group.

Hence we also have as a particular case

Theorem D. *Let M be a hyperbolic 3-manifold with cusps and H a finitely generated prosoluble subgroup of the profinite completion of $\pi_1(M)$. Then one of the following statements holds:*

(i) H is virtually a subgroup of a free prosoluble product of abelian pro-p groups;

(ii) H is a virtually abelian.

7 The fundamental group of standard compact arithmetic manifolds

In this section we give a direct application of Proposition 6.4.

Let $\mathrm{SO}(n,1)$ be the special orthogonal group and R_k the ring of integers of an algebraic number field k. Recall that a hyperbolic n-manifold is a manifold $M^n = \mathbb{H}^n/\Gamma$ such that Γ is a torsion-free subgroup of $O(n,1)$.

Definition 7.1. A *standard arithmetic subgroup* Γ of $\mathrm{SO}(n,1)$ is a group commensurable with the subgroup $\mathrm{SO}(q,R_k)$ of R_k-points of

$$\mathrm{SO}(q) = \{X \in \mathrm{SL}(n+1,\mathbb{C}) : X^t q X = q\}.$$

An arithmetic hyperbolic n-manifold $M^n = \mathbb{H}^n/\Gamma$ such that Γ is standard arithmetic is a *standard arithmetic manifold.*

It is shown in [BHW11] that standard uniform arithmetic subgroups of $\mathrm{SO}(n,1)$ are virtually compact special and in [PS24, Lemma 6.3 and Theorem 7.4] this was extended to standard non-uniform arithmetic subgroups. Also, it is well-known that the fundamental group of these manifolds is hyperbolic relative to virtually abelian subgroups. Thus, Theorems B and C can be applied to standard arithmetic manifolds.

Theorem 7.2. *Let M be a standard arithmetic manifold and H a finitely generated prosoluble subgroup of the profinite completion of $\pi_1(M)$. One of the following holds:*

(i) H is a subgroup of a free prosoluble product of abelian pro-p groups;

(ii) H is abelian.

Moreover, if $\pi_1(M)$ is uniform, then H is projective.

8 The fundamental group of compact 3-manifolds

To prove Theorem E we will use the pro-p version of it, i.e., the classification of pro-p subgroups of the profinite completion of a 3-manifold group established by Henry Wilton and the second author in [WZ18, Theorem 1.3].

Theorem 8.1. *[WZ18, Theorem 1.3] A finitely generated pro-p subgroup of the profinite completion of the fundamental group of a compact, orientable 3-manifold M is a free pro-p product of pro-p groups from the following list of isomorphism types:*

(i) For $p > 3$: C_p; $\mathbb{Z}_p$; $\mathbb{Z}_p \times \mathbb{Z}_p$; the pro-p completion of $(\mathbb{Z} \times \mathbb{Z}) \rtimes \mathbb{Z}$ and the pro-p completion of a residually-p fundamental group of a non-compact Seifert fibred manifold with hyperbolic base of orbifold;

(ii) For $p = 3$: in addition to the list of (i) we have a torsion-free extension of $\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3$ by C_3;

(iii) For $p = 2$: in addition to the list of (i) we have C_{2^m}; D_{2^k}; Q_{2^n}; $\mathbb{Z}_2 \rtimes C_2$; the torsion-free extensions of $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ by one of C_2, C_4, C_8, D_2, D_4, D_8, Q_{16}; the pro-2 extension of the Klein-bottle group $\mathbb{Z} \rtimes \mathbb{Z}$; the pro-2 completion of all torsion-free extensions of a soluble group $(\mathbb{Z} \rtimes \mathbb{Z}) \rtimes \mathbb{Z}$ with a group of order at most 2.

Now we are ready to prove the theorem that classifies prosoluble subgroups of the profinite completion of 3-manifold groups.

Theorem E. *Let M be a compact orientable 3-manifold. If H is a finitely generated prosoluble subgroup of $\widehat{\pi_1(M)}$, then one of the following statements holds:*

(i) H is isomorphic to a subgroup of a free prosoluble product of pro-p groups from the following list of isomorphism types:

(1) For $p > 3$: C_p; $\mathbb{Z}_p$; $\mathbb{Z}_p \times \mathbb{Z}_p$; the pro-p completion of $(\mathbb{Z} \times \mathbb{Z}) \rtimes \mathbb{Z}$ and the pro-p completion of a residually-p fundamental group of a non-compact Seifert fibred manifold with hyperbolic base of orbifold;

(2) For $p = 3$: in addition to the list of (1) we have a torsion-free extension of $\mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_3$ by C_3;

(3) For $p = 2$: in addition to the list of (1) we have C_{2^m}; D_{2^k}; Q_{2^n}; $\mathbb{Z}_2 \rtimes C_2$; the torsion-free extensions of $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2$ by one of C_2, C_4, C_8, D_2, D_4, D_8, Q_{16}; the pro-2 completion of the Klein-bottle group $\mathbb{Z} \rtimes \mathbb{Z}$; the pro-2 completion of all torsion-free extensions of a soluble group $(\mathbb{Z} \rtimes \mathbb{Z}) \rtimes \mathbb{Z}$ with a group of order at most 2;

(ii) $H \simeq \widehat{\mathbb{Z}}_\pi \rtimes C$ is a profinite Frobenius group where C is a finite cyclic group and π is set of primes;

(iii) H is a subgroup of a central extension of either D_{2n} or tetrahedral group T or octahedral group O by a cyclic group of even order;

(iv) H is a subgroup of the profinite completion of a 3-dimensional Bieberbach group B (i.e, B is torsion-free virtually $\mathbb{Z}^3$);

(v) H is a subgroup of the profinite completion of a group extension of $\mathbb{Z}^2 \rtimes_A \mathbb{Z}$ by C where C is trivial or C_2 and $A \in \mathrm{GL}_2(\mathbb{Z})$ (and A is an Anosov matrix if C is trivial), or a central extension of $\mathbb{Z}$ by $\mathbb{Z}^2$;

(vi) H is an extension of a torsion-free procyclic group $\widehat{\mathbb{Z}}_\sigma$ by a subgroup H_0 of a free prosoluble product of finitely many finite cyclic p-groups with H_0 acting either trivially on $\widehat{\mathbb{Z}}_\sigma$ or by inversion.

8.1 Reduction to the irreducible case

Let M be a compact, orientable, non-geometric 3-manifold. Since every compact manifold is a retract of a closed manifold, it follows easily that we may reduce to the case in which M is closed.

If M is reducible, by the Prime Decomposition Theorem, we can write

$$\pi_1(M) = \pi_1(M_1) * \cdots * \pi_1(M_n)$$

where each M_i is a prime 3-manifold. A prime 3-manifold is either irreducible or $S^2 \times S^1$ or the non-orientable S^2 bundle over S^1. In the last two cases, the fundamental group is isomorphic to $\mathbb{Z}$ (see [Hem04] and [GSW21, Page 5]).

Thus applying [Zal24, Theorem 1.4] we obtain

Theorem 8.2. *Let H be a finitely generated prosoluble subgroup of the profinite completion $G = \widehat{\pi_1(M)}$ of a reducible 3-manifold having a non-trivial free product decomposition*

$$\pi_1(M) = \pi_1(M_1) * \cdots * \pi_1(M_n) * F$$

where each M_i is a irreducible 3-manifold and F is a free group of finite rank. Then one of the following holds:

(i) all intersections $H \cap \widehat{\pi_1(M_i)}^g$, $g \in G$ are pro-p for some fixed prime p and $\sum_v d(H \cap \widehat{\pi_1(M_i)}^g) \leqslant d(H)$;

(ii) $H \simeq \widehat{\mathbb{Z}}_\pi \rtimes C$ is a profinite Frobenius group;

(iii) $H \leqslant \widehat{\pi_1(M_i)}^g$ for some $g \in G$.

In case (i) we apply Theorem 8.1 to $H \cap \widehat{\pi_1(M_i)}^g$ to deduce the structure of each of these intersections and obtain (i) of Theorem E. Case (ii) of the theorem above corresponds to (ii) of Theorem E.

Thus we only need to consider (iii) and in this case we may assume, w.l.o.g, that $H \leqslant \widehat{\pi_1(M_i)}$, i.e., we may assume that M is irreducible.

8.2 Reduction to the geometric cases

Now we can use the following:

Proposition 8.3 ([WZ18], Proposition 4.1)**.** *Let M be a closed irreducible 3-manifold. Then the action of $\widehat{\pi_1(M)}$ on the standard profinite tree S associated to the JSJ-decomposition of M is 4-acylindrical.*

Then we can apply Theorem F to deduce:

Theorem 8.4. *Let M be a closed irreducible 3-manifold. If H is a finitely generated prosoluble subgroup of $\widehat{\pi_1(M)}$, then H embeds into a free profinite product*

$$\coprod_{v \in V} H_v$$

of H-stabilizers of vertices in S (the standard profinite tree associated to $\widehat{\pi_1(M)}$) and one of the following statements holds:

(i) *H is isomorphic to a subgroup of a free prosoluble product of finitely many finitely generated pro-p groups $H \cap uH_v u^{-1}$ for every $v \in V$ and $u \in \coprod_{v \in V} H_v$;*

(ii) *H is a profinite Frobenius group $\widehat{\mathbb{Z}}_\pi \rtimes C$;*

(iii) *H is a subgroup of H_v for some $v \in V$.*

If H satisfies (i), then we can apply Theorem 8.1 again and deduce (i) or (ii) of Theorem E. The case (ii) corresponds to Theorem E (ii). Therefore we may assume that H satisfies item (iii) and since H in this case is conjugate into a vertex of $\widehat{\pi_1(M)}$ we may assume that H is contained in the profinite completion of a geometric manifold.

Thus, to complete the proof of Theorem E we need to study the profinite completion of $\pi_1(M)$ when M is geometric.

8.3 Geometric cases

The Geometrization conjecture, proved by G. Perelman on his approach to prove the Poincaré conjecture (see [Per02], [Per03b], [Per03a]), says that a 3-manifold is geometrically modeled by one of the eight Thurston's geometries

(i) S^3,	(iii) $\mathbb{R}^3$,	(v) Sol	(vii) $\mathbb{H}^2 \times \mathbb{R}$,
(ii) $S^2 \times \mathbb{R}$,	(iv) Nil,	(vi) $\mathbb{H}^3$,	(viii) $\widetilde{\mathrm{SL}_2(\mathbb{R})}$.

The fundamental group of the first five are, respectively, of the following types:

(i) a finite cyclic group or a central extension of D_{2n}, T, O or I by a cyclic group of even order. We are denoting by T the tetrahedral group of 12 rotational symmetries of a tetrahedron, by O the octahedral group of 24 rotational symmetries of a cube or of an octahedron, by I the icosahedral group of 60 rotational symmetries of a dodecahedron or an icosahedron;

(ii) isomorphic to $\mathbb{Z}$ or the infinite dihedral group;

(iii) virtually abelian;

(iv) virtually nilpotent;

(v) soluble.

The fundamental groups in the case (i) are soluble. except the case of an extension of I by a cyclic group of even order. The icosahedral group I is isomorphic to A_5, which is non-soluble. Recall that A_5 has no proper subgroups of order greater than 12 so the possible subgroups are C_2, C_3, C_5, D_3, D_5, A_4 and they are covered in the previous cases. Thus, to analyze the soluble subgroups when Γ is a central extension of A_5 by C_{2n} we can reduce it to the other soluble cases.

If we consider H as a prosoluble subgroup of the profinite completion of groups (i)-(v), then H can be any of its subgroups. We summarize the existing classification as:

(i) H is a finite cyclic group or H is a subgroup of a central extension of a dihedral, tetrahedral, octahedral or icosahedral group by a cyclic group of even order;

(ii) H is a subgroup of the profinite completion of the infinite dihedral group;

(iii) H is a subgroup of the profinite completion of a 3-dimensional Bieberbach group (i.e, virtually $\mathbb{Z}^3$);

(iv) H is a subgroup of the profinite completion of a group extension of $\mathbb{Z}^2 \rtimes_A \mathbb{Z}$ by C where C is trivial or isomorphic to C_2 and $A \in \mathrm{GL}_2(\mathbb{Z})$ or a central extension of $\mathbb{Z}$ by $\mathbb{Z}^2$;

The reader can check [AFW15], [Sco83, Section 4], [Ner21, Section 1.5, Chapter 2, Chapter 3] and [GM23] for details on the fundamental groups of the first five items.

The item (vi) (hyperbolic case) is the subject of Theorem A. Since a finitely generated projective profinite group is a subgroup of a non dihedral free profinite product (cf. Proposition 2.12) it is a particular case of Theorem E, item (i). So, it remains to deal with cases (vii) and $(viii)$.

8.4 Seifert manifolds

In order to conclude the analysis of the possible irreducible compact 3-manifolds we now deal with the Seifert manifolds. We have the following exact sequence:

$$1 \to \mathbb{Z} \to \pi_1(M) \to \pi_1(O) \to 1$$

where $\pi_1(O)$ is a Fuchsian group acting on $\mathbb{Z}$. Thus we describe first prosoluble subgroups of a profinite completion of a Fuchsian group.

Theorem 8.5. *Let H be a prosoluble subgroup of a profinite completion of a Fuchsian group $\widehat{\pi_1(O)}$. Then one of the following holds:*

(i) H is a subgroup of a free prosoluble product of finitely many finite cyclic p-groups;

(ii) $H \simeq \widehat{\mathbb{Z}}_\pi \rtimes C$ is a profinite Frobenius group;

(iii) H is a finite cyclic group.

Proof. Passing to an open subgroup containing H if necessary (see [HKS71]), we can replace O by a non-trivial finite-sheeted cover of degree $n > 2$ whose profinite completion contains H, hence, we can assume that $\pi_1(O)$ is not a triangle group. In this case we have

$$\widehat{\pi_1(O)} = \widehat{K} \amalg_{\widehat{C}} \widehat{F}$$

where K is a free product of cyclic groups, F a free profinite group of finite rank and C an infinite cyclic group (see [LMR96]).

Since $\widehat{C}$ is malnormal in $\widehat{\pi_1(O)}$, by [WZ17, Lemma 7.3], $\widehat{\pi_1(O)}$ is a 1-acylindrical finite graph of profinite groups. Applying Theorem F we get one of the following cases:

(a) H is a subgroup of the free prosoluble product of intersections $H \cap \widehat{F}^h$, $h \in \widehat{\pi_1(O)}$ and $H \cap \widehat{K}^g$, $g \in \widehat{\pi_1(O)}$, and these intersections are pro-p.

(b) H is a profinite Frobenius group.

(c) H is contained in either $\widehat{F}$ or $\widehat{K}$ up to conjugation.

Suppose that (a) holds. Note that $H \cap \widehat{F}^g$ is free pro-p and $H \cap \widehat{K}^g$ is free pro-p product of cyclic pro-p groups by the pro-p version of the Kurosh Subgroup Theorem (cf. [Rib17, Theorem 9.6.2]). Then we have (i). If H is Frobenius, we have item (ii).

Suppose that (c) holds. If H is the conjugate of $\widehat{F}$ then it is projective and so (i) holds. Otherwise we may assume that $H \leqslant \widehat{K}$. Then applying Theorem F again we deduce the statement of the theorem. $\square$

Theorem 8.6. *Let G be the fundamental group of a Seifert manifold with either geometry $\mathbb{H}^2 \times \mathbb{R}$ or $\widetilde{\mathrm{SL}_2(\mathbb{R})}$ and H a finitely generated prosoluble subgroup of $\widehat{G}$. Then H is an extension of a torsion-free procyclic group $\widehat{\mathbb{Z}}_\sigma$ by a subgroup of a free prosoluble product of finitely many finite cyclic p-groups.*

Proof. We have the following exact sequence:

$$1 \to \mathbb{Z} \to \pi_1(M) \to \pi_1(O) \to 1$$

where $\pi_1(O)$ is a Fuchsian group acting on $\mathbb{Z}$ either trivially or by inversion. Then we have an action of $\widehat{\pi_1(O)}$ and hence of $H_0 = H\widehat{\mathbb{Z}}/\widehat{\mathbb{Z}}$ on $\widehat{\mathbb{Z}}$ and this action is trivial or by inversion.

Thus we have an exact sequence

$$1 \to \widehat{\mathbb{Z}}_\sigma \to H \to H_0 \to 1,$$

where $\widehat{\mathbb{Z}}_\sigma = H \cap \widehat{\mathbb{Z}}$ and H_0 acts on $\widehat{\mathbb{Z}}_\sigma$ either trivially or by inversion.

By Theorem 8.5 we have the following 3 possibilities for H_0.

(a) H_0 is a subgroup of a free prosoluble product of finitely many finite cyclic p-groups;

(b) $H_0 \simeq \widehat{\mathbb{Z}}_\pi \rtimes C$ is a profinite Frobenius group;

(c) H_0 is a finite cyclic group.

and we analyze each of them in turn.

(a) In this case H is an extension of $\widehat{\mathbb{Z}}_\sigma$ by H_0 with the either trivial action or by inversion.

(b) Suppose that
$$1 \to \widehat{\mathbb{Z}}_\sigma \to H \to \widehat{\mathbb{Z}}_\pi \rtimes C_n \to 1$$
is an exact sequence with $\widehat{\mathbb{Z}}_\pi \rtimes C_n$ a profinite Frobenius group with C_n a finite cyclic group of order n. As the lift of every element of finite order of $\pi_1(O)$ to $\pi_1(M)$ must centralize $\mathbb{Z}$, the lift of every element of finite order of H_0 must centralize $\widehat{\mathbb{Z}}_\sigma$. But H_0 is generated by torsion elements, so the extension is central in this case.

Let ρ be the set of all primes dividing n. Then the preimage of C_n in H is $\widehat{\mathbb{Z}}_{\rho'} \times \widehat{\mathbb{Z}}_\rho$ where ρ' is the complement of ρ in σ (it is isomorphic but not equal to $\widehat{\mathbb{Z}}_\sigma$). Thus we can write H as $(\widehat{\mathbb{Z}}_{\rho'} \times \widehat{\mathbb{Z}}_\pi) \rtimes \widehat{\mathbb{Z}}_\rho \cong \widehat{\mathbb{Z}}_{\rho'} \times (\widehat{\mathbb{Z}}_\pi \rtimes \widehat{\mathbb{Z}}_\rho))$. As $\pi \cap \rho = \varnothing$, $\widehat{\mathbb{Z}}_\pi \rtimes \widehat{\mathbb{Z}}_\rho$ is projective and hence a subgroup of a free profinite product of cyclic p-groups (since a finitely generated projective group is a subgroup of a free group of finite rank).

(c) Suppose that
$$1 \to \widehat{\mathbb{Z}}_\sigma \to H \to C_n \to 1$$

is an exact sequence with C_n finite cyclic of order n. As the lift of every element of finite order of $\pi_1(O)$ to $\pi_1(M)$ must centralize $\mathbb{Z}$, the lift of every element of finite order of H_0 must centralize $\widehat{\mathbb{Z}}_\sigma$. Hence the extension is central and so H is procyclic torsion-free.

□

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